# Colour-blind Series: Protanopia

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## What is it for?

Protanopia is a form of colour blindness that affects the ability to see red. People with protanopia cannot distinguish between red and green, and red tones can appear very dark or almost black. It's one of two types of colour blindness that fall under the 'red-green colour blind' umbrella.

This matters in digital products, because colour is often used to share important information. A form that highlights errors in red, or a chart using red and green bars side by side, becomes very difficult to interpret.

## How to support it

The key principle is that colour should always be paired with another visual cue.

- Designers can add icons, labels, or patterns next to colour to show meaning. They can also use colour blind simulation tools to check designs before they are built
- Developers can ensure interactive states like errors and warnings are distinguishable by more than just colour (e.g. by using icons or supporting text)
- Content writers can make sure any meaning shown through colour is also stated in words

## Useful Links

WCAG 1.4.1: Use of Colour

Colour Blindness Simulator (Coblis)

Colour and Accessibility (WebAIM)